

The Burden of the Broken Grid: Modelling power-sector reliability to support low carbon development in Nigeria

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ABSTRACT

Nigeria has one of the greatest electricity deficits globally and, even in areas connected to the central grid, struggles to provide reliable power across the nation. Frequent system collapses and widespread reliance on diesel generation present a burden for Nigerian households and the economy as a whole. One causal factor in these collapses is capacity inadequacy owing to reduced plant availability as plants are frequently non-operational due to maintenance or other management issues. Using a combination of OSeMOSYS and FlexTool modelling, this study shows the significant burden that persistently unavailable plants present for decarbonisation of the Nigerian energy system. Modelling which includes reliability improvements nearly halves total system costs and emissions versus business-as-usual. Further, Nigeria is unable to meet its 2021 Nationally Determined Contributions (NDCs) without such improvements, indicating that increasing plant availability and reducing diesel generator use must be prioritized in policy to support national implementation of these targets.

1. Introduction

While Nigeria has the largest population of any African nation, the available generation capacity for the country sits far below that of many other nations, at just 6.7 GW reliably connected to the grid [1]. This capacity is dominated by gas power plants which are frequently out of service due to maintenance or fuel constraints [2]. The unreliability of the fleet and the grid itself mean that the nation experiences frequent blackouts [3,4]. Approximately 60.5 % of the population has access to electricity with that number falling to 27 % in rural areas [5]. For many Nigerians, the lack of reliable grid power means that backup generators are a part of daily life: over 86 % of businesses and nearly a quarter of households own a generator [6,7]. Persistent system failures and the prevalence of diesel generation come with significant development impacts, as power system inadequacy is estimated to create annual GDP losses of 6 % or USD 22.8 Billion [8].

On top of the on-grid emissions from fossil fuel generation, emissions from diesel generation are likely to be a serious hurdle in meeting the country's latest Nationally Determined Contribution (NDC) [9]. Further, the unreliability of the current generation fleet, if not addressed, raises questions about the flexibility of the system. This will determine its ability to handle the increasing penetration of variable renewable

energy (VRE) needed to meet the NDC power system decarbonisation target.

While other studies have modelled the Nigerian energy system, none have yet evaluated the country's ability to meet the latest round of NDCs. Further, in the body of published work there is little optimisation modelling that explores diesel generation or the persistent issues with plant availability. The flexibility of the system given these variables is also yet to be explored.

This leads to a key, as yet unanswered, question: how will Nigeria balance its grid reliability issues and the resulting reliance on diesel generation with its climate commitments? This study aims to answer that question using energy systems modelling to explore the impacts of improved plant availability on alternative pathways for the development of Nigeria's power sector. Given the context presented above (and further explored in Section 2), these impacts primarily centre around system adequacy and emissions/costs of diesel generator use. Ultimately, this analysis will synthesize these impacts and explore their implications for meeting the nation's development targets, including the latest NDC.

The current power system in Nigeria is described in the first part of Section 2, while the remainder of the section is devoted to a discussion of previous modelling work. Section 3 describes the methodology used in this study, including any assumptions and data used in the modelling.

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Acronyms

OSeMOSYS	The Open Source Energy Modelling System
NDC	Nationally Determined Contribution
UNDC	Unconditional Nationally Determined Contribution
CNDC	Conditional Nationally Determined Contribution
VRE	Variable renewable energy
BAU	Business-as-usual
EE	Energy efficiency
T&D	Transmission and distribution
IRENA	International Renewable Energy Agency
CCGTs	Combined cycle gas turbine
SCGT	Single cycle gas turbine

Section 4 presents the results from both the OSeMOSYS and FlexTool models, which show that increased plant availability reduces loss of load and consequently is anticipated to reduce generator usage. Further, reduced reliance on self-generation is found to nearly halve emissions and system costs. Section 5 compares the results to previous work and analyses the feasibility of improvements assumed in the modelling, along with the implications of each development pathway for meeting NDC targets. Finally, Section 6 summarises key findings and implications for policymaking.

2. Background

2.1. Nigeria's current power system

Nigeria's power system is dominated by gas power plants which make up 80 % of the on-grid generation capacity, with the remaining 20 % made up of hydropower. While these plants represent 13,014 MW of generation capacity, actual generation has never surpassed 6000 MW [10]. The generation shortfall can be largely attributed to low plant availability –capacity is considered unavailable when it cannot be used to reliably generate power. Availability can be constrained by a variety of factors, including lack of preventative maintenance, inadequate gas supply to plants, water shortages, vandalism, and transmission constraints [2].

Loss of plants to maintenance issues cannot solely be attributed to the age of the fleet as Nigeria's gas plants are an average of 12.6 years old, approximately halfway through their lifespan [11]. Further, the 2019 available capacity of the five newest gas plants in Nigeria (added between 2015 and 2018) is 64 % of their rated capacity. This is not far above the overall fleet average of 43 % and far below the international standard [1,12]. Surveys of gas plants in the nation largely point to poor maintenance culture, insufficient supply of spare parts, and funding issues as causal factors for plant downtime [12]. In terms of options for improvement, regular preventative maintenance of plants as well as improved tariff structures to reinforce cost-effective practices may help to target these issues [3,12].

The historic underinvestment and corruption in sector after privatisation have contributed to these significant operating deficits [13,14]. Reviewing the arrangements in the sector, Roy et al. find that there is “networked corruption” in the system where all actors are benefiting in some way from corruption creating little impetus for reform [14]. These political and economic hurdles compound the technical challenges of maintaining a functioning power system.

These plant and grid management issues result in semi-frequent system collapses; there were four instances of voltage collapse in 2022 and a trend of ten or more such collapses per year in the last decade [15, 16]. Given the frequency of blackouts, backup diesel-powered generation has emerged as a key resource for maintaining access to electricity in both the industrial and residential sectors. Estimates indicate that

over 86 % of businesses and almost a quarter of homes have generators and that installed capacity is between 15 and 20 GW [6,7]. These estimates put the generator capacity equal to or nearly double the installed on-grid capacity.

Due to high diesel prices, electricity produced by these diesel generators is twice as expensive as power supplied by the grid [17]. Reliance on self-generation in the industrial sector is estimated to add 30 % of the price of domestic products over imports [18]. Failing to address system management issues in the nation will reinforce this reliance on self-generation and continue to weigh down Nigerian homes and businesses with the cost of supplying their own electricity.

The use of diesel generators also creates emissions with significant climate and health impacts for the nation. One study estimated that diesel generation in Nigeria “produces CO₂ emissions equivalent to 60 % of its annual electric sector emissions” [19]. This is largely attributed to the low efficiency of generators when compared to traditional power plants. In sub-Saharan Africa, emissions from diesel generation also likely account for the “majority of power sector emissions of nitrogen oxides (NO_x) and fine particulate matter (PM_{2.5})” as generators produce higher amounts of these pollutants per kWh than many typical generation sources [7]. Due to the proximity of generators to homes, the high PM emissions are particularly worrisome for local air quality and the health.

Recent policy commitments have looked to address this issue, including the 2022 Energy Transition Plan which stresses diesel self-generation as an “urgent priority for decarbonisation” [20]. The plan further calls for the elimination of diesel generation by 2050 to support a 2060 net zero target.

Nigeria has also signed on to the Paris Agreement and has committed to decarbonisation in the long term. The 2021 update to their conditional and unconditional NDCs reiterated this commitment and further reduced their emissions ceiling. These updates incorporated more pollutants in the emissions calculations and used a less aggressive growth rate, in-line with observed growth from 2015 to 2021 [9]. The new growth rate led to lower BAU emissions projections (452 MT versus 898 MT), but the nation has maintained its unconditional commitment of 20 % below BAU by 2030. The conditional commitment has been increased to 47 % below BAU (up from 45 %). Given lower BAU estimates, these updated commitments represent lower actual emissions limits. These goals present ambitious green development targets requiring significant investment and thoughtful implementation. Out of the 2021 NDCs, only the conditional NDC addresses the issue of backup diesel generation directly. Even then, the target simply specifies the “elimination of diesel self-generation by 2030” without any supporting objectives [9].

2.2. Previous modelling of Nigeria's power system

Several studies have modelled the Nigerian energy system in the last decade through a broad range of lenses. Given the electricity access and grid issues in the nation, a number have focused on the “electricity gap” and the nation's ability to meet growing demand. Avila et al. [21], Ibrahim and Kirkil [22], and Dada and Moser [23] all explore this issue, but their scenarios have an explicit focus on on-grid technologies and do not consider the current or anticipated future role of self-generation in the energy mix. Their modelling anticipates that the expansion of on-grid capacity can meet growing demand but do not address plant availability. These studies also all appear to use current installed capacity as the baseline in models. As addressed above, even newer plants are subject to availability issues and the expansion of capacity alone is unlikely to solve reliability issues in the nation.

Daggash and Mac Dowell (2021) found that Nigeria would be able to deliver its initial NDC targets with no added costs versus business-as-usual (BAU). This study used highly granular data and was able to model regional grid connections which provided further insight into the political and economic feasibility of their scenarios. The scenarios modelled did include significant amounts of diesel in the initial model

period but diesel fades from the generation mix by 2030 and appears again later in the model period. From the modelling it is unclear if improvement in plant performance is the reason for this decrease or if capacity expansion alone makes up for increased on-grid generation.

This existing modelling of the Nigerian system is interested in the energy gap and in energy efficiency but does not specifically focus on the issue of self-generation in the absence of reliable on-grid power. When diesel generation has been explored, it has mostly been explored in the context of an off-grid technology rather than explicitly connected to the issue of plant availability and the reliability of the power grid.

3. Methodology

This study aims to model this dynamic and how it may impact the newest round of NDC commitments which have also yet to be modelled. Further, few studies have incorporated flexibility modelling for the country, and developing such a model with an open-source tool like FlexTool can serve as an important stepping stone for future work. The tools used for this modelling and any assumptions made are outlined below.

3.1. OSeMOSYS and FlexTool

OSeMOSYS is a bottom-up energy systems model which optimises system costs for a given set of technologies and demands. The set of demands, fuels, and technologies used in this study are outlined in the reference energy system presented in Fig. 1. As shown, end-use demands in this study are not exogenously split between on and off grid, rather the model may choose to expand the grid when cost-effective versus off-grid generation.

The OSeMOSYS dataset used in this study is built from the base file described in Allington et al. (2021) with some adjustments and further assumptions outlined below.

Given the focus on system reliability, OSeMOSYS is limited in its representation of time, the model used here simplifies a year into four seasons and eight timeslices to represent a day and night in each season. To overcome this issue and to add depth to the analysis, OSeMOSYS results were put into the International Renewable Energy Agency's FlexTool to further analyse the hourly flexibility and adequacy of the resulting generation mix.

FlexTool is another open-source tool which models an energy system's ability to meet hourly demand each year. Key inputs include reserve margin as well as capacities and technical data of generating technologies. The model then outputs the anticipated loss of load, capacity inadequacy, and VRE curtailment [24]. Here, the model is used to validate the OSeMOSYS scenarios and to explore the flexibility tradeoffs associated with persistent plant availability issues.

3.2. OSeMOSYS inputs: current power system data

As outlined before, Nigeria's power system is dominated by gas power plants with a few large hydropower projects. Active plants were identified using historic power sector data from the Nigerian Electricity Regulatory Commission [1,11,25]. Table 1 shows the installed capacity of each technology in 2021.

Power plants and renewables projects currently under construction were also included in the residual capacity in the model and have been set to start generating in line with expected commissioning dates as per the Global Energy Monitor [26]. The investment costs for these projects are not included in the costs for each scenario to highlight the investment needed to fill the gap in generation in the optimisation modelling over and above any capacity currently under construction.

The exact installed diesel generation capacity in Nigeria remains unknown, but estimates project that approximately 8–14 GW were in use in 2016, 10–15 GW in 2017, and 2019 capacity was in the range of 15–20 GW [7,27,28]. To estimate the existing capacity used in this

study, the medians of these estimates were used for their respective years and the average growth between estimates was used to estimate capacity in the remaining years. After initial model runs, capacity was refined to fill unmet demand given historical generation data from 2015 to 2021.

Generation of on-grid technologies is limited by this same generation data from the first seven years in the modelling period (2015–2021). During this initial model period, investment in transmission capacity is also constrained to current and historic capacity.

3.3. OSeMOSYS inputs: techno-economic assumptions

Other key techno-economic assumptions include a discount rate of 10 % and reserve margin of 15 %, consistent with previous modelling studies [2,29]. Costs, technology lifetimes, and capacity factors were taken from Allington et al. (2021) [30]. Renewable energy potentials, and fuel reserves were also taken from this dataset with the exception of hydropower potential which used a conservative estimate which considered only operational and planned hydropower projects as opposed to the technical potential.¹

To minimise flexibility issues on the already fragile grid, utility-scale VRE generation was limited across all scenarios and across the entire modelling horizon to 30 % of demand each year with 20 % coming from wind and 10 % coming from solar. This cap is in line with IRENA modelling of the West African Power Pool which used the same penetration caps to ensure system reliability across scenarios [31].

Across all scenarios, the build-out of nuclear is constrained by a construction lead time (no plant can begin operation until 2029) and an annual build rate cap of 0.5 GW. This build rate is consistent with previous work by Daggash & Mac Dowell (2021) who used this rate across their modelling to capture the existing issues with expanding grid capacity while allowing the model to invest as necessary to roughly meet demand [2].

3.4. Scenarios

Four base scenarios were modelled (see Table 2) to explore the impacts of BAU practices and pathways to meet Nigeria's 2021 NDC commitments.

Where scenarios include gradual investment in energy efficiency, annual investment across the modelling horizon is constrained to 5 % of the 2050 capacity deployed in a run without any efficiency constraint specified in Cannone et al. to ensure that investment is reasonable [32].² Beyond the constraints outlined above, investment in other technologies is determined by the cost-optimisation in the modelling.

3.5. Sensitivity analysis

Each scenario was run with three different timescales for plant availability improvement associated with improved management practices: improvement to 2030, 2050, and no improvement. The primary assumption here is that any load not met by on-grid plants due to low plant availability will be met by diesel generation. The availability factors by technology can be found below in Table 3. These factors represent the percentage of the installed capacity of a technology that can be reliably called upon to generate power. Downtime due to maintenance and other miscellaneous work is accounted for in each technology's capacity factor as specified in Allington et al. which have been taken from regional average values. For CCGTs and SCGTs, this value is 85 % and for other fossil plants this value is 80 % [30]. In practice, this

¹ Further information on the input data used in modelling be found in the supplementary data files.

² For additional information on these constraints see the supplementary data files.

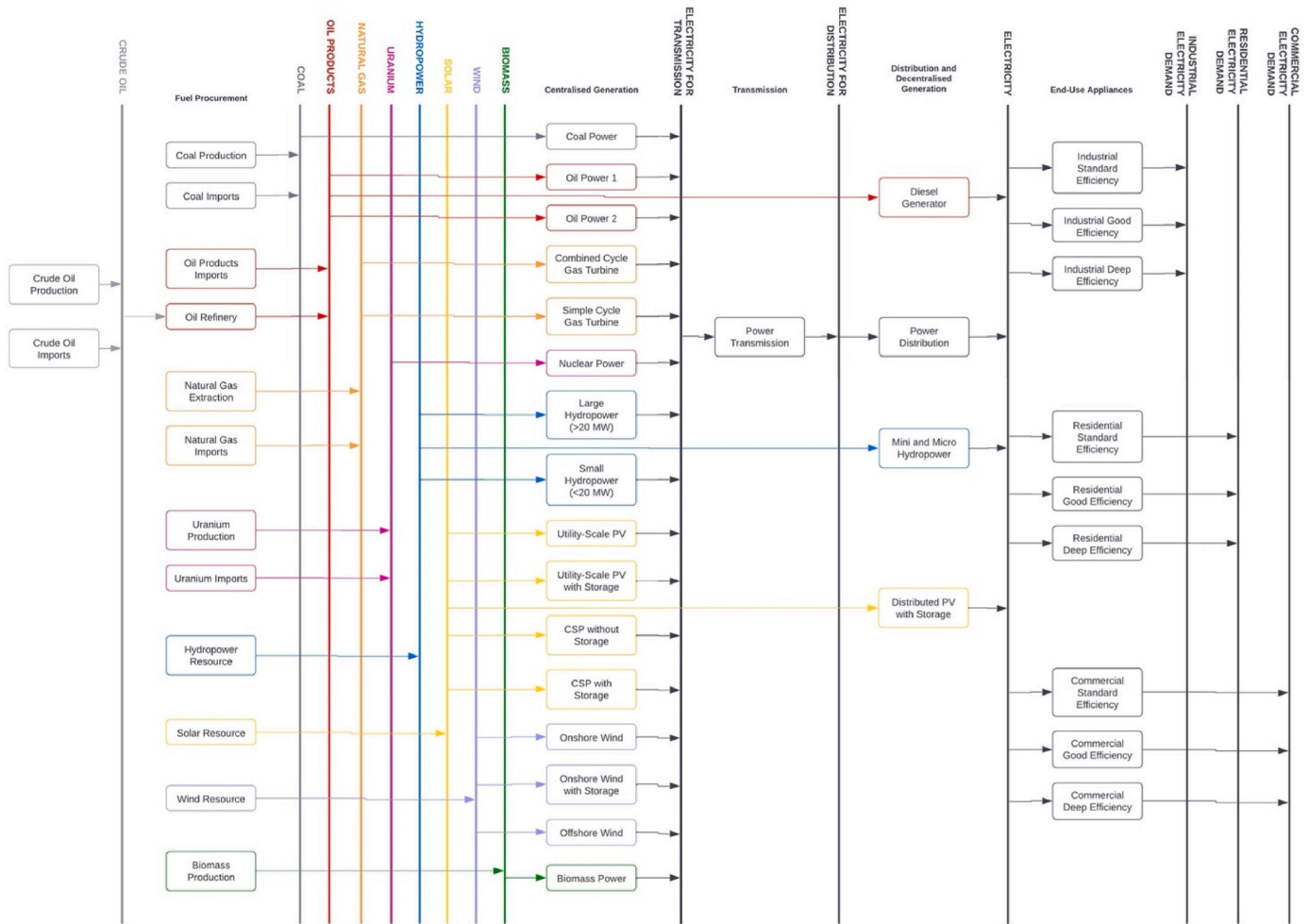


Fig. 1. Reference energy system for Nigeria. Adapted from Allington et al. (2021). CSP = Concentrating Solar Power.

Table 1

2021 Installed generation capacity. Data from NERC (2018, 2021) and GEM (GEM, 2022).

Technology	2021 Installed Capacity (GW)
Gas (CCGT)	4.738
Gas (SCGT)	6.721
Large Hydropower (>20 MW)	1.99
Diesel	17.9
Off-Grid Solar PV	0.0264
Micro Hydro (<1 MW)	0.0004

means that the availability factor models the gap between the current performance of generators in Nigeria and a reasonable uptime in the region which still sits below industry best practice of 95 % [12]. In scenarios with no improvement, generators will never meet this threshold of availability but in scenarios with improvement to 2030 and 2050 plants will reach this regional average availability in 2030 and 2050 respectively.

This variation creates the 12 model runs discussed below.

3.6. Flexibility analysis: FlexTool

This study uses a single-node FlexTool model with primary inputs

Table 2

Core scenarios used in modelling.

Scenario	Description and Constraints
BAU	On-grid investment is constrained to only natural gas and hydropower. Transmission and distribution (T&D) losses improve gradually from a base of 15 % in 2018, reaching 13 % by 2030. No investment in energy efficiency (EE).
Least Cost (LC)	Minimal constraints to find least-cost solution. No investment in nuclear power but investment in other technologies is unconstrained. T&D losses improve gradually, reaching 13 % by 2030. Gradual investment in EE is permitted.
Unconditional NDC (UNDC)	Emissions capped at 20 % below BAU emissions in 2030 (85.3 MT) from 2030 onwards. T&D losses improve gradually, reaching 13 % by 2030. Gradual investment in EE is permitted.
Conditional NDC (CNDC)	Emissions capped at 47 % below BAU emissions in 2030 (56.5 MT) from 2030 onwards. All SCGTs are phased out by 2030. 13 GW off-grid renewables, 3.5 GW of small hydro, and 12 GW of new large hydro capacity added by 2030. Investment in other technologies is unconstrained. T&D losses improve to international standard (8 %) by 2030. Gradual investment in EE is permitted.

Table 3

Availability factors. Data calculated using NERC (2021, 2018).

	Baseline Availability (no improvement)	Annual Improvement to 2030	Annual Improvement to 2050
Combined Cycle Gas Turbine	40 %	8 %	2 %
Single Cycle Gas Turbine	47 %	7 %	2 %
Other Fossil	50 %	6 %	2 %
On-Grid Hydropower	68 %	4 %	1 %
Other Renewables	80 %	3 %	1 %

derived from the OSeMOSYS modelling. The reserve margin (15 %) and costs and efficiencies of technologies are the same as was used for the OSeMOSYS modelling (sourced from Allington et al., 2021).³ The updated demand for scenarios with EE and the capacity of each technology were taken directly from the results. Only the available capacity for each model run was added to FlexTool and diesel capacity was not included as FlexTool focuses on-grid generation and diesel capacity is assumed to be used for backup generation. As FlexTool explores the flexibility of a system over a single year, the flexibility analysis presented here focuses on model year 2040.

To feed into the higher temporal resolution FlexTool modelling, hourly country-specific capacity factors for variable renewable generation and hourly demand profiles were taken from the PLEXOS-World 2015 dataset [33]. Many technical constraints for the different generation sources in the modelling are not included in the OSeMOSYS model but were added to the FlexTool modelling to account for the varied technical flexibility potential of each technology. Some of these parameters (minimum load, ramp rates, minimum downtime/uptime and inertia constants) could not be sourced from Nigeria-specific datasets and so global averages have been used.³ The value of lost load used in the flexibility analysis was set at \$10,000 USD per MWh which is consistent with the value used by Daggash and Mac Dowell (2021) and typical values used in FlexTool [2,34].

3.7. Limitations

The primary limitation across the modelling was limited access to Nigeria-specific data notably in terms of historic plant availability and plant operation costs. This meant that it was not possible to build the anticipated costs of availability improvement directly into the model. Future work could explicitly model costs of improvement, but a full energy systems model may not be the most suitable avenue for this work. As specified above, some technical parameters in the FlexTool and OSeMOSYS modelling could also not be sourced from Nigeria-specific studies and datasets. Where possible, averages for Africa have been used, but some global averages have been included where no regional data was available.

In both the OSeMOSYS modelling and in a single-node model in FlexTool there is additionally a simplification of the geographic complexity of the grid. In OSeMOSYS, the lack of spatially resolved costs means that the modelling may not accurately reflect the costs of grid expansion which can vary significantly by location. In the FlexTool modelling, a single-node model fails to account for any potential grid congestion or the role of the geographic dispersion of renewables.

The OSeMOSYS and FlexTool modelling presented here is also focused solely on the power system and decarbonisation within that sector. The decarbonisation of road transport, industry, and cooking are

all additional decarbonisation challenges for the nation but are not within the scope of this analysis and are not expected to be electrified in the short to medium term. Expanding this analysis to explore the whole energy sector would more accurately capture the demands expected to shift to the power sector later in the model period as heating and transport are electrified. This additional electrification would also allow for the exploration of sector coupling in the flexibility analysis. Further flexibility work could also incorporate regional transmission to shed greater light on the role of the current transmission grid in system reliability issues and what investments may be needed as VRE penetration increases.

4. Results and analysis

Results from modelling the Nigerian energy system are presented below from 2015 to 2065 with the model period running until 2070 to account for end-game optimisation issues at the end of the model period which are typical of bottom-up optimisation models. Out of the twelve model runs, three runs produced a generation shortfall, deploying the model's backstop. This backstop is included as a very expensive "dummy" technology with unconstrained capacity that the model can deploy if it is unable to meet demand in any scenario to keep the model from failing. Table 4 shows the runs with backstops and the size of the shortfall in TWh. To contextualise these figures, the current (2022) electricity demand is 81 TWh and is expected to rise to 919 TWh in 2065 [35]. Across the model horizon, the 2314 TWh of energy generated by the backstop in the UNDC scenario with no plant availability improvement is 14 % of the total energy generated in that scenario from 2015 to 2065. This shortfall is additional to the shortfall met by diesel generation and is due to emissions constraints in NDC scenarios.

Across scenarios, the generation results presented below show that increased plant availability decreases the use of diesel generators. Increased availability also reduces total system costs and emissions; however, the pace of the improvement (to 2030 versus 2050) has a much smaller impact on both costs and emissions. However, modelling also suggest that in the near term it is possible for Nigeria to meet its NDCs with little or no change to its generation mix if plant availability is improved.

4.1. Generation

4.1.1. Business-as-usual

As Fig. 2 shows, capacity is dominated by CCGTs in the model runs with plant improvement to 2030 and 2050. SCGTs, which are the current dominant form of turbine, are phased out after 2040 in all scenarios. In the scenarios with improvement in plant availability, diesel generation is phased out once plants reach full availability. Without improvement, diesel remains an important part of the generation mix (30–50 % of generation) throughout the model period.

Hydropower does not come close to its technical potential of 64.3 TWh of annual generation, and peaks at 15 TWh of generation from in 2030. Decentralised PV takes over 4 % of generation in 2050 in all scenarios and capacity peaks in 2062 with 22.875 GW installed across model runs.

As no investment in EE is enabled in BAU, variation in the total amount of electricity generated is due to the mix of on/off grid generation — scenarios with high amounts of diesel need to generate less as less energy is lost to the low T&D efficiency which starts at 15 % in this

Table 4

Generation shortfall in model runs across the model horizon (2015–2065).

	BAU	Least Cost	UNDC	CNDC
Improvement to 2030	OK	OK	OK	OK
Improvement to 2050	OK	OK	14 TWh	OK
BAU/No Improvement	OK	OK	2314 TWh	1381 TWh

³ Additional details on the input data used in the FlexTool modelling can be found in the supplementary data files.

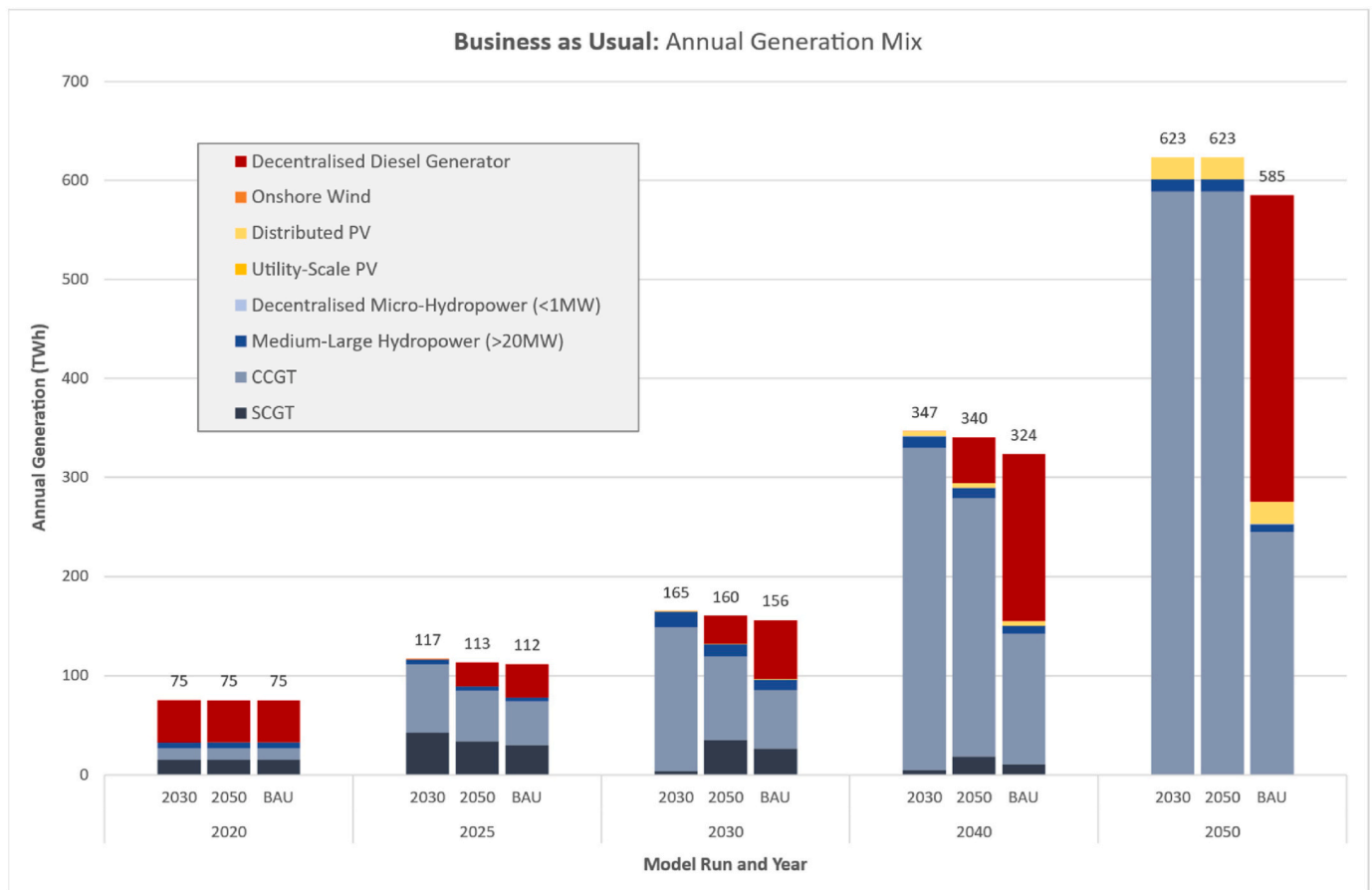


Fig. 2. Generation in the business-as-usual scenario.

scenario and gradually improves to 13 % in 2030.

4.1.2. Least cost

The least cost generation mix, shown in Fig. 3, also includes high amounts of natural gas which is largely made up of CCGTs with SCGT capacity declining after 2025. Biomass also plays a significant role from 2035 onwards and ultimately makes up for 27 % of power generated in 2065 in most runs. Coal enters the generation mix after 2025 and remains a stable part of the generation mix until the end of the model period.

The renewable energy mix includes similar amounts of hydropower to the BAU scenario. Onshore wind also plays a small role in the generation mix with utility-scale PV supplying 10 % of power generated in 2050 across most model runs.

Diesel generation varies significantly across model runs with higher rates of diesel in runs without availability improvement. Greater variation in the generation mixes and investment in EE mean that diesel generation is lower in Least Cost model runs than in the BAU runs. As was seen in the BAU model, in runs with availability improvement diesel generation is phased out before plant availability reaches target levels.

4.1.3. UNDC

Only one UNDC run, the run with availability improvement to 2030, could successfully meet demand without surpassing emissions targets (see Fig. 4). In all other runs, the model's backstop kicked in indicating that no technologies available would be able to meet demand given the constraints on the model. The results for these "failed" runs are presented but the backstops are indicated, and the power generated by backstops is assumed to come from diesel self-generation.

CCGTs remain an important part of the generation mix in the UNDC

model runs with SCGTs playing smaller role and disappearing from the mix after 2040. Coal is the only other centralised fossil resource in the model runs but coal generation is roughly half of what it was in the Least Cost runs.

The model invests aggressively in nuclear from 2029 onwards and this makes up 16 % of generation in 2050 across runs. Hydropower hits its investment ceiling in 2054. On-grid solar hits its production limit in 2040 and wind hits its limit in 2050. Off-grid PV is largely absent across model runs until 2050 where it supplies 4–5 % of electricity generated.

Even in scenarios without improvement, diesel generation makes up less of the annual power generated in UNDC model runs than in Least Cost or BAU runs.

4.1.4. CNDC

Only two CNDC runs were able to run without triggering the model's backstops — the runs with availability improvement to 2050 and 2030. It is likely that more of the CNDC runs were successful than the UNDC runs (despite the lower emissions limit) due to the forced investment early in the model period in off-grid renewable energy which lessens the on-grid load.

As per scenario constraints and NDC targets, SCGTs phase out of generation by 2030 but CCGTs generate throughout the model period. As seen in Fig. 5 coal does enter the generation mix in 2030 but at half the level observed in UNDC runs. As was observed in UNDC model runs, nuclear is deployed to the fullest extent permitted as are wind and solar.

Across all models runs, the level of diesel generation is slightly below the corresponding run's generation in the UNDC by an average of 3 %.

4.1.5. All scenarios

Fig. 6 highlights the generation mix across the scenarios in 2040, the

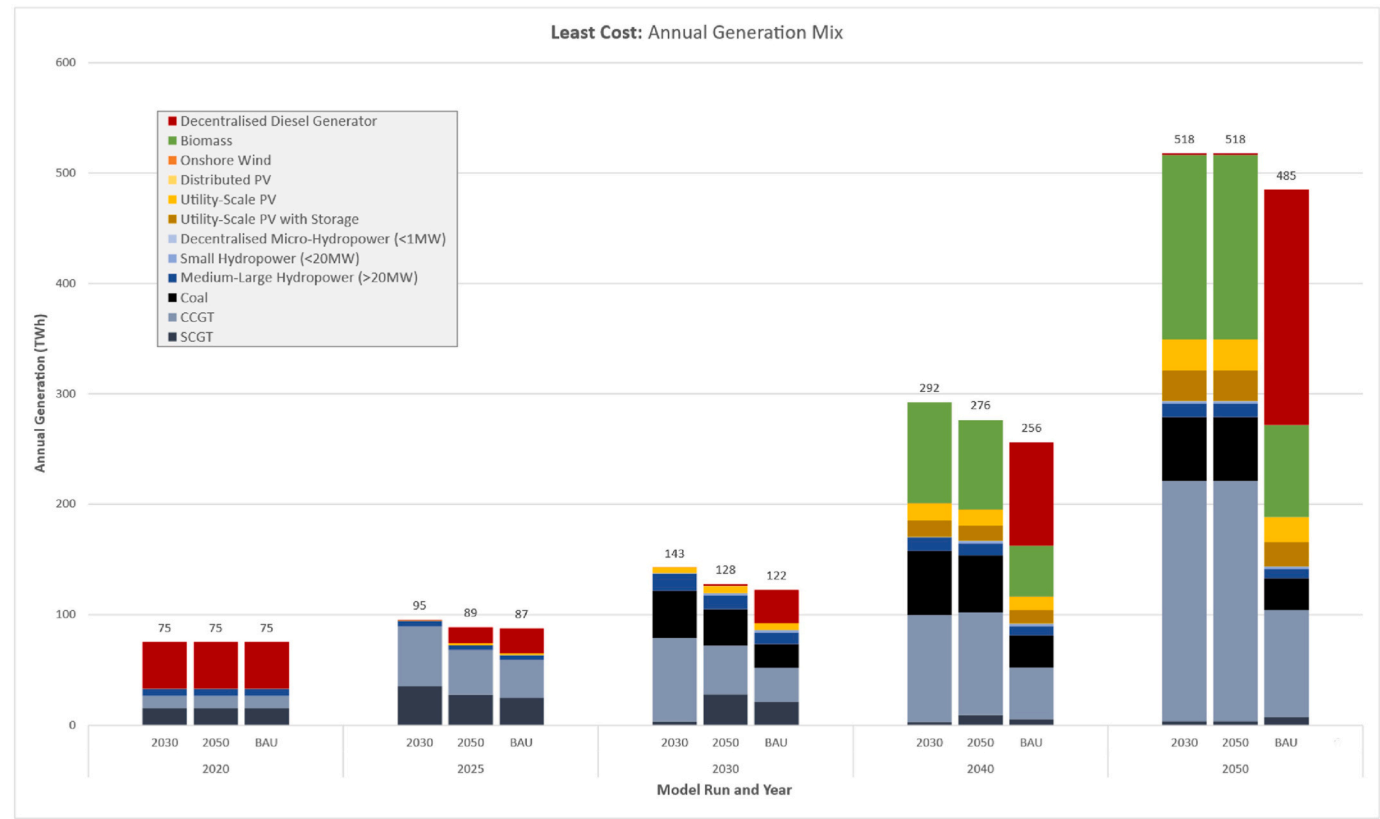


Fig. 3. Generation in the Least Cost scenario.

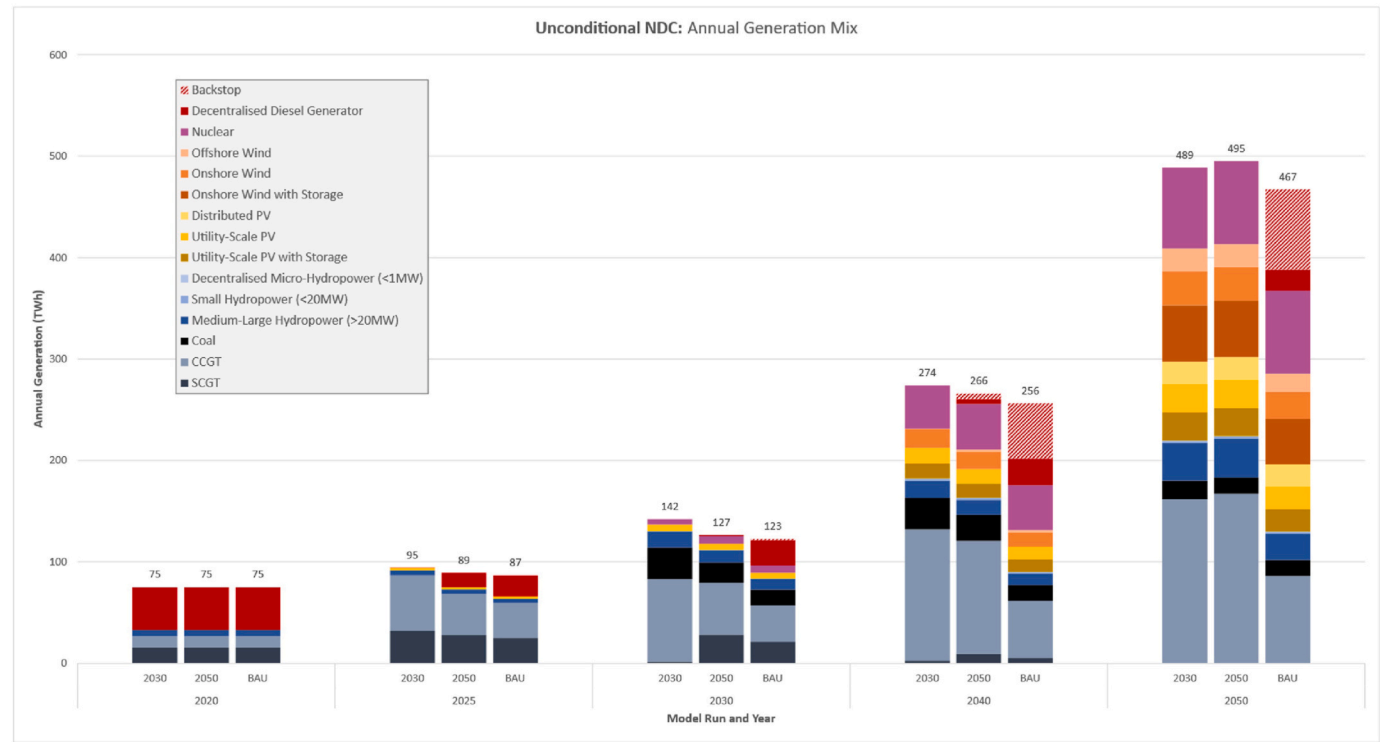


Fig. 4. Generation in the UNDC scenario.

year studied in the flexibility analysis in FlexTool. Across all scenarios and sensitivities there are large shares of CCGT generation. There is also significant reliance on diesel generation in model runs with no plant

availability improvement. NDC scenarios include nuclear capacity which is absent from the Least Cost and BAU generation mix, for the NDC scenarios this low-carbon baseload is needed to meet the ambitious

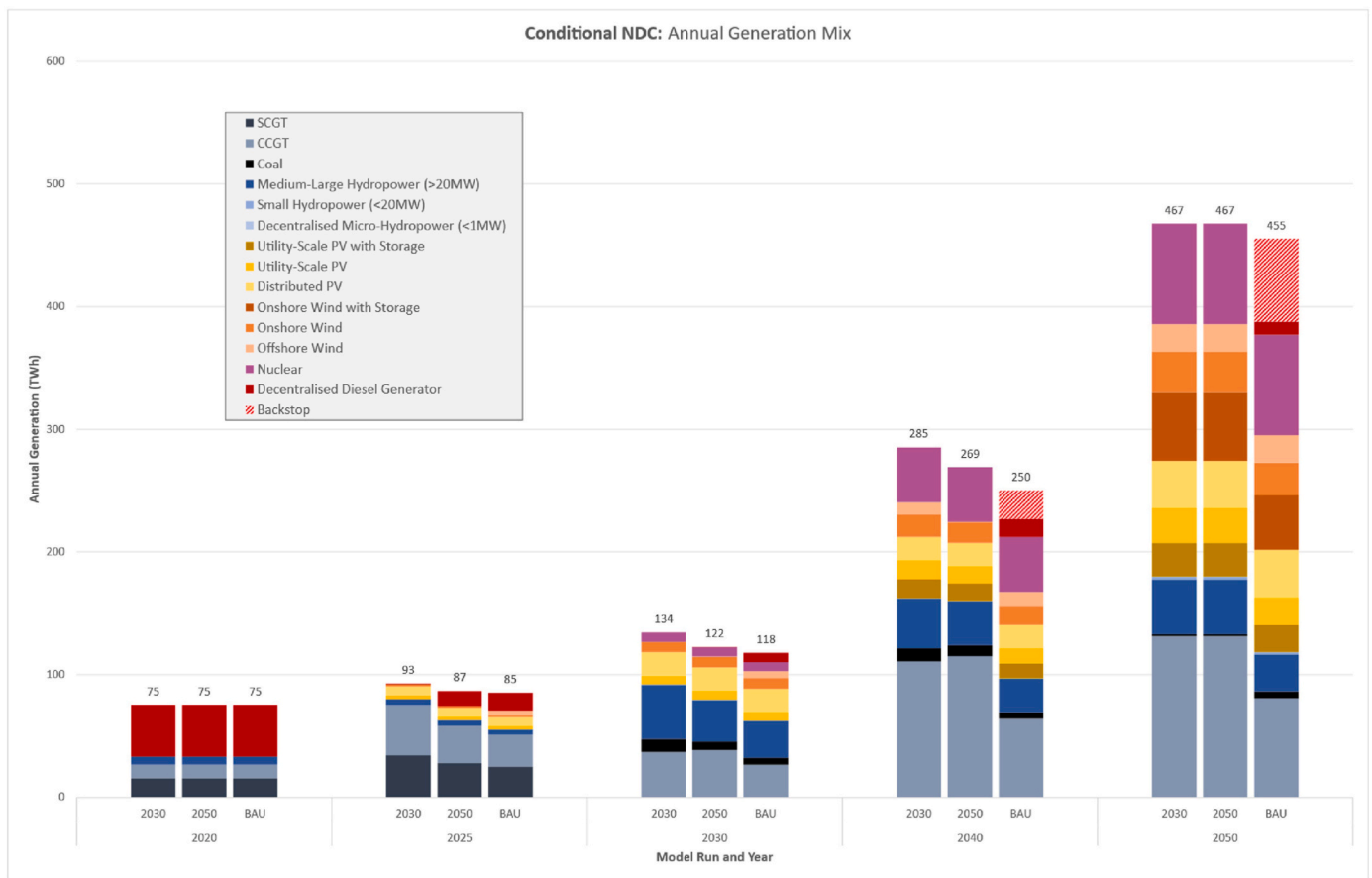


Fig. 5. Generation in the CNDC scenario.

power sector decarbonisation targets. The Least Cost generation mix is also the only set of scenarios to include biomass generation.

As explained above, variation in the total volume of energy generated across scenarios is due to the different levels of energy efficiency deployed and the varied split between on-grid and off-grid generation.

4.2. Emissions and NDC targets

As shown in Fig. 7, the BAU model run without availability improvement has the highest emissions overall, followed by the Least Cost run without availability improvement. The UNDC and CNDC model runs have the lowest total emissions over the model period, even when energy generated by the backstop is assumed to be generated by diesel generators.

Across all scenarios, runs with no availability improvement have consistently higher emissions – an average of 45 % above those with improvement across scenarios. There are also smaller differences in emissions between runs with 2030 improvement versus runs with 2050 improvement. A slower rate of plant improvement increases total emissions by an average of 2 % over the model period.

Comparing emissions over time, most scenarios have consistent emissions (within 100 MT CO₂e of each other) until 2034 where results begin to diverge more dramatically. In the period of 2030–2035 the Least Cost run without availability improvement surpasses the annual emissions of the BAU scenarios with improvement. In 2065, all Least Cost runs have annual emissions higher than the two BAU model runs with availability improvement. This difference between the Least Cost and BAU scenarios is likely due to the deployment of biomass in the Least Cost runs after 2035.

Despite these changes, later in the model period all Least Cost runs with availability improvement are able to achieve the 2030 UNDC

target, as shown in Fig. 8. The BAU run with full plant availability in 2030 also remains below the emissions threshold. Notably, no viable model runs without availability improvements can hit the NDC targets. The CNDC model runs without improvement meet the targets, but these results are qualified by the fact that these savings early in the model period are likely reflective of insufficient investment, as scenarios are not able to meet demand later in the model period.

4.3. System costs

Fig. 9 shows the total discounted system costs from 2022 to 2065 across scenarios and model runs. Runs with backstops activated do not have costs shown as the high cost of the backstop technologies skews the costs in the run. Costs include all CAPEX, fuel costs, operational costs, costs of improved energy efficiency, and T&D expansion costs. Notably, the costs of improved plant performance are not included.

As expected, the Least Cost scenarios have the lowest discounted total system costs when compared with the relevant corresponding runs of other scenarios. The cheapest scenario overall is the Least Cost run with availability improvement to 2030 – at USD 149 billion – as this scenario has the least constrictive constraints.

Across scenarios, system costs decrease by an average of 45 % as plant availability improves gradually to 2050. This equates to average savings of USD 138 billion over the modelling period. The difference between runs with improvement to 2030 versus 2050 is much lower – a slower rate of improvement only raises system costs by an average of 14 %.

4.4. Flexibility

Across scenarios, increased available capacity reduced loss of load.

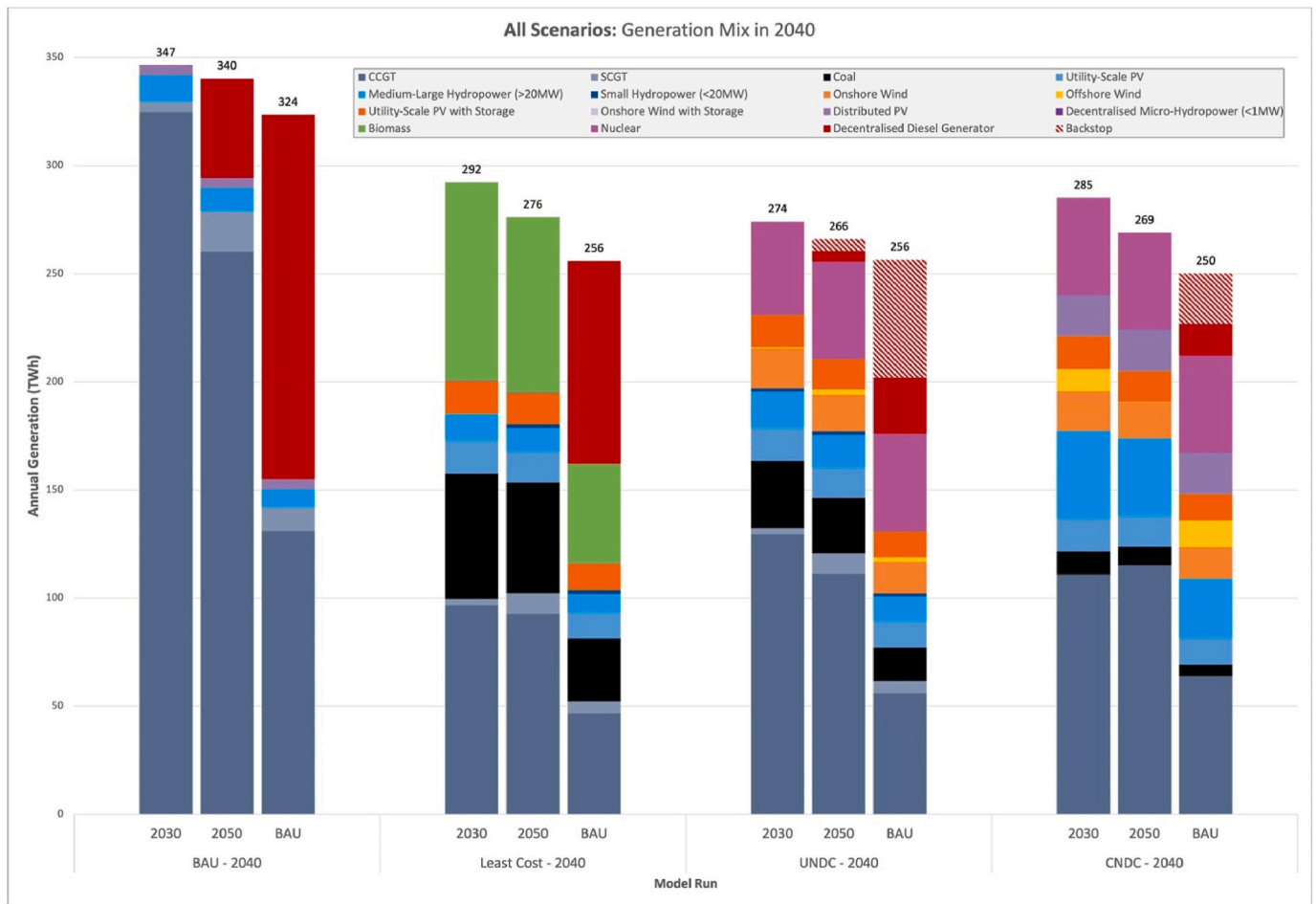


Fig. 6. Generation mix across scenarios in 2040.

The BAU run without availability improvement had the highest loss of load in 2040, with 41 % of demand not met. The three other runs without improvement had the next highest losses as shown in Table 5.

Some loss of load comes from ramp rate limitations in the modelled generation mix, but most of the losses appear to be caused by a lack of capacity. This inadequacy can be seen in Table 5 which shows key flexibility indicators, including the peak net load for the year and available capacity. The peak net load is the demand less the amount of VRE generation, and the available dispatchable capacity is the capacity that can be used to meet the remaining demand. This net load is 10–20 GW more than the available capacity in some scenarios.

In all scenarios other than BAU, the annual lost load in FlexTool is greater than the diesel generation in the OSeMOSYS modelling by 2–10 TWh. This indicates that the OSeMOSYS modelling does slightly underestimate the supply gap filled by self-generation. Despite this difference, the results from the two models are broadly consistent – improved availability does improve the system adequacy (as shown in the FlexTool results) and reduces the reliance on diesel generation, building a more resilient system.

Likely due to the cap on VRE, there was minimal curtailment across runs. The CNDC run had the highest curtailment with 0.3–0.7 % of VRE curtailed. Given this VRE limit and use of natural gas in all scenarios, the inertia across scenarios is also not an issue, with zero instances of insufficient inertia in any scenario.

Broadly, the results of the flexibility analysis support the OSeMOSYS modelling in stressing the importance of improved availability for system reliability. The NDC runs with full plant availability in 2030 have some loss of load in the analysis which would need to be met by diesel or increased on-grid investments not accounted for in the OSeMOSYS

modelling. This indicates that improved availability is unlikely to resolve all flexibility issues on its own, and therefore greater investment in storage or sector coupling (which are outside the scope of this study) may be needed to fully optimise the system. Future work could explore this dynamic.

5. Discussion

5.1. Comparisons to previous modelling

The two NDC scenarios presented above were fairly diversified with wind and solar meeting 30 % of demand, bolstered by gas, hydro, and nuclear power. Daggash and Mac Dowell's (2021) modelling of Nigeria's first set of NDCs produced a similar generation mix overall – large amounts of gas with some hydro, coal, solar PV, wind, and nuclear. By 2030, their scenarios have no diesel in the generation mix which matches the runs with 2030 improvement, but runs with slower/no improvement do have diesel generation in 2030. Dioha and Kumar's (2020) NDC modelling does not include any wind or nuclear and instead has much more solar throughout the modelling period. The diesel generation in their modelling for 2030 varies from 25 to 50 TWh. The model run for the UNDC without plant availability improvement explored here sits at the low range of their modelling at 26 TWh coming from diesel and backstops.

Another key finding in the modelling presented in this paper was that Nigeria would be able to meet its 2030 NDC targets without altering its generation mix if plant availability is aggressively improved. Daggash and Mac Dowell (2021) came to a similar conclusion that the 2015 NDCs could be met without overhauling the generation mix. Their scenarios

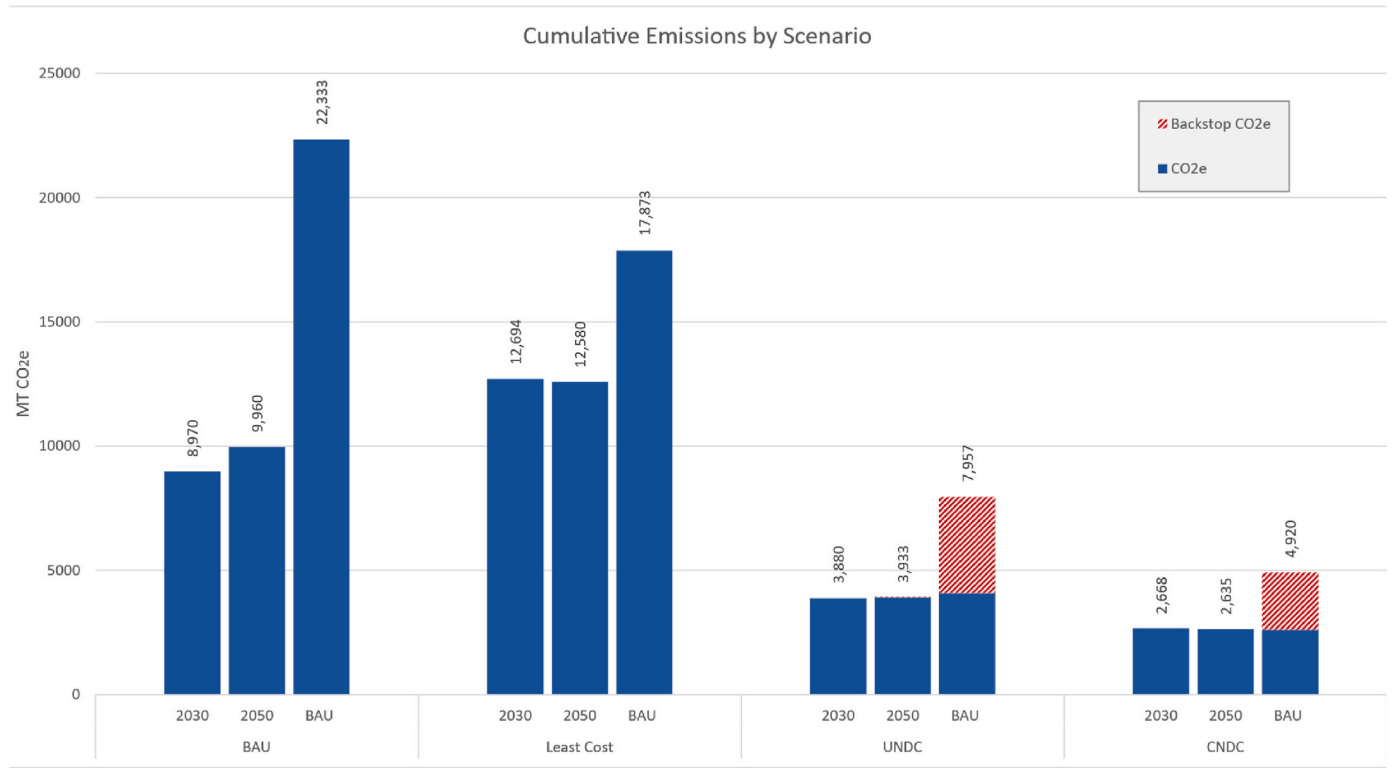


Fig. 7. Cumulative emissions by scenario from 2015 to 2065.

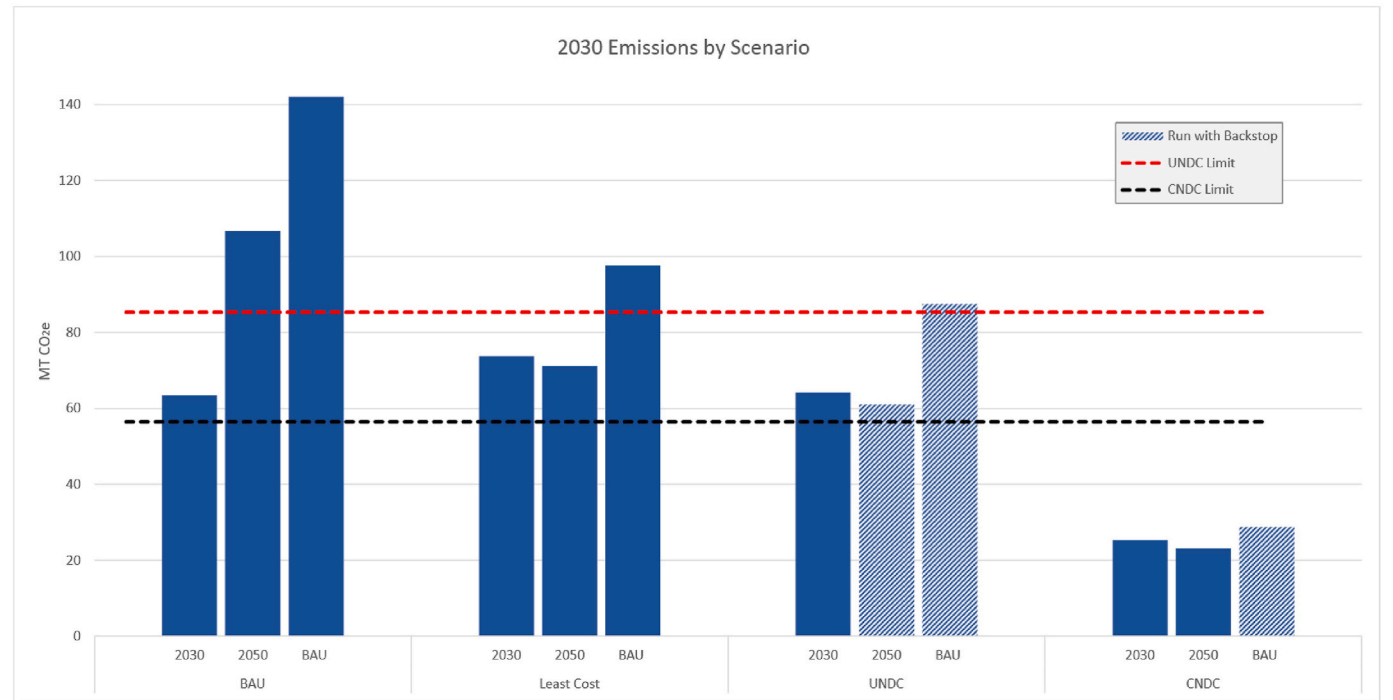


Fig. 8. 2030 emissions by scenario.

can be considered comparable to the scenarios presented here with full plant availability, which also meet the updated 2021 NDCs.

5.2. Feasibility

Taking the importance of increasing plant reliability for reducing

emissions, it is important to contextualise the feasibility of both the proposed rates of plant improvement and the costs associated with implementation.

The model runs with improvement to 2030 anticipate that gas plant availability will improve by 7–8 % annually, while runs with 2050 improvement have a lower rate of improvement at only 2 %. Historic

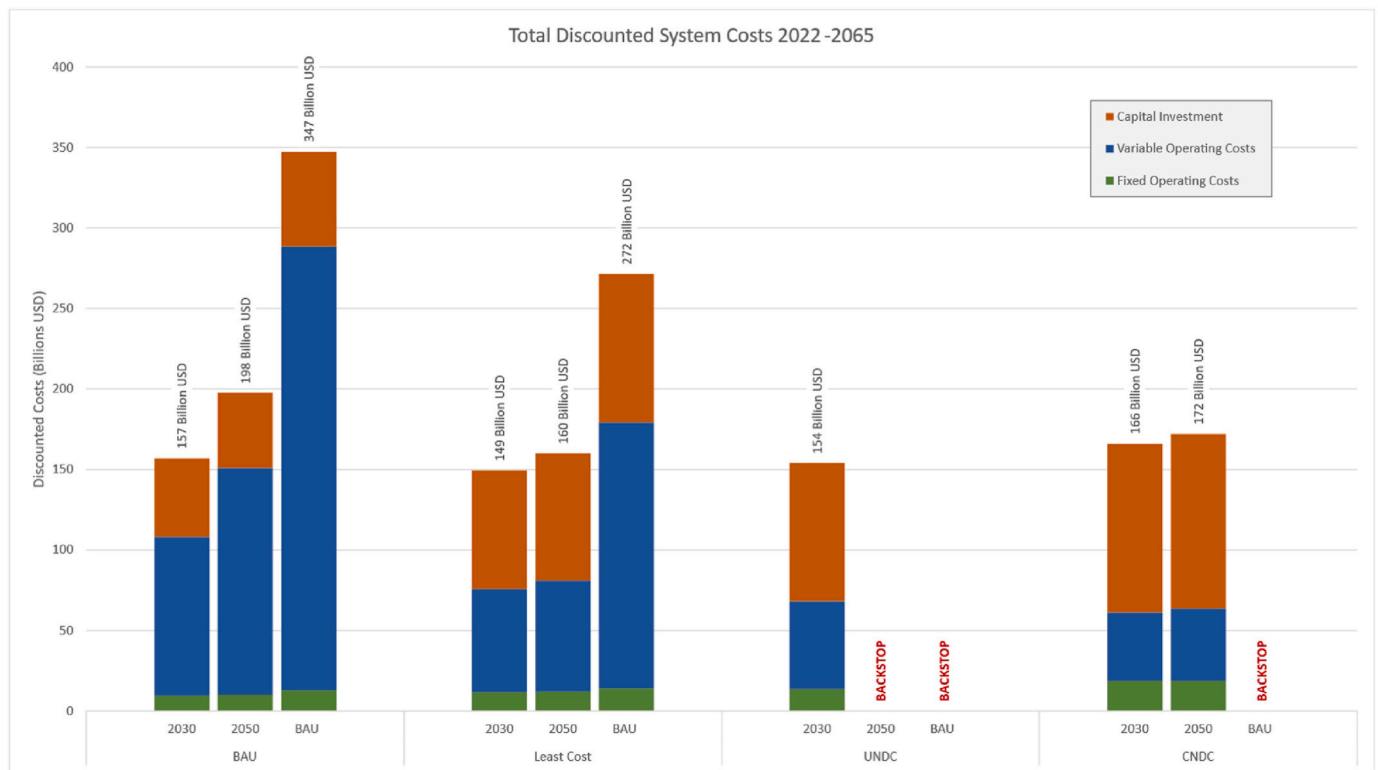


Fig. 9. Total discounted system costs from 2022 to 2065 by scenario.

Table 5
Key flexibility metrics.

	Loss of load (% of demand)	Insufficient reserves (% of reserve)	Peak net load (GW)	Available Dispatchable Capacity (GW)
BAU with Improvement to 2030	0.04	7.402	48.54	51.58
BAU with Improvement to 2050	3.39	31.15	48.70	40.93
BAU with No Improvement	41.05	99.63	48.99	21.75
LC with Improvement to 2030	0	0	40.65	49.24
LC with Improvement to 2050	0.07	7.801	38.25	41.46
LC with No Improvement	10.72	68.14	37.46	24.72
UNDC with Improvement to 2030	1.00	14.35	35.12	36.41
UNDC with Improvement to 2050	3.15	20.06	34.13	31.07
UNDC with No Improvement	15.53	89.07	34.56	20.69
CNDC with Improvement to 2030	0.83	13.31	35.23	43.62
CNDC with Improvement to 2050	3.53	20.21	34.52	36.30
CNDC with No Improvement	9.53	52.38	31.72	24.82

Entries in red indicate instances where the peak net load is greater than the available dispatchable capacity, leading to loss of load.

data on plant availability in Nigeria are limited, but a 2011 report does place average gas plant availability at 40 %, which is 3 % lower than the 2019 availability figures used as the baseline for this study⁴ [1,36]. This would indicate an average improvement of 0.4 % a year over the 8-year period. Since 2011, however, there have been significant changes in the power sector which include privatisation of distribution and generation in 2013. This transition has been far from smooth and, after nearly a decade, there are still significant issues with tariff structures in the new market [3]. Privatisation likely had a negative impact on plant availability as new management struggled in untested waters. While the 2050 model runs have higher rates of improvement than the historic progress (2 %) there is hope that a decade on from privatisation that more

proactive plant management can help improve reliability in the sector. In this context, the 2 % annual improvement may be achievable, while 7–8 % – as given in the improvement to 2030 model runs – may be an overambitious target. The modelling above shows that the higher rate of improvement (runs with improvement to 2030) does offer added emissions reductions, but those reductions are minimal over the model period when compared to runs with slower improvement to 2050.

The above modelling does not include any costs for the increased plant maintenance and management needed to make availability improvements. There are limited data on the anticipated costs for improving plant availability in the literature. Given this, the decision was made to not include any added cost for availability improvement. Instead, the cost difference between scenarios with improvement can be used to gauge a ceiling below which availability improvement would make economic sense. If the costs of improvement are below the difference between a run with improvement and a run without, then the cost of the run with improvement accounting will be economic.

Runs without improvement are between USD 111 to 190 billion more expensive than runs with improvement, as costs nearly double in BAU and LC runs without improvement. It seems unlikely that gradual availability improvement would double system costs, given that some costs could logically be recouped through increased power generation per unit. In this context, while the exact costs of improved management are murky, it seems that they are likely lower than the costs of continued mismanagement even without accounting for the value of the lost load or the social and environmental costs of generator usage, which would further increase the expense of runs without improvement.

5.3. NDCs and policy implications

In the context of the nation's NDCs, results indicate that Nigeria will need improvement in plant availability/management to meet any commitments. Aggressively improving plant availability would allow the nation to hit its 2030 UNDC target without changing the generation mix. A more gradual, and likely more feasible, improvement pathway

⁴ Here an average of the SCGT and CCGT availability factors used here (43.5 %) is compared to the number used in the 2011 study as they did not indicate the split between CCGT and SCGT plants in the installed capacity.

would require more aggressive investment in solar PV and some deployment of EE, as demonstrated in the Least Cost run with 2050 improvement. This timescale of eliminating diesel generation by 2050 is also aligned with the target set out in the Energy Transition Plan [20].

In the long term, improved availability will also become important for dispatchable generation as the share of variable renewable energy (VRE) is raised over 30 %. The low flexibility of runs with reduced availability, even at low levels of VRE, indicate that a system with such low available capacity is unlikely to be able to respond to even quicker and more dramatic changes in net load that are expected at higher VRE penetration. This is particularly important given the announcement that the nation intends to reach net zero in 2060 [37].

While the broader global warming implications of generator use are vital, local particulate matter and air quality implications also provide a strong argument for improving Nigeria's energy sector. In the BAU scenario, generator use will increase ten-fold by 2065. The negative health impacts are already apparent with current levels of generator usage and are likely to increase with rising generator usage. There are also significant added costs to homes, businesses, and the economy that come with this continuing reliance. These added costs have significant consequences for sustainable development in the nation and would present significant hurdles in reaching SDG goals 3 (Good Health and Well-being), 7 (Affordable and Clean Energy), and 13 (Climate Action) most directly [38].

Given these findings, the prioritisation of plant maintenance and availability improvement is particularly important in national policy to support the 2021 NDC commitments as long-term decarbonisation goals cannot be met without improvements in plant availability. To target this issue, previous work has suggested the introduction of preventative maintenance practices and improved tariff structures which incentivise similar cost-effective operational measures [3,12]. Given the political context in the country and the historic underinvestment and corruption which have led to this level under-performance, improvements must also be coupled with an infusion of capital alongside meaningful anti-corruption reforms and a rebuilding of social trust in the sector [13, 14].

Beyond the NDCs, other literature has explored the role of distributed PV and storage as a viable zero-emissions alternative to off-grid diesel generation [7,39]. Investigating the viability of such systems in Lagos, Babajide and Brito find that PV systems are a feasible alternative to diesel generation. While the higher upfront costs may still present a limitation, these systems are shown to provide a cost savings of 60 % over their lifetime on top of the emissions reductions delivered [39]. The success of low carbon energy communities globally could also be applied to this context as an opportunity to build social trust in the sector while centring low carbon sources of energy [40].

5.4. Regional and global relevance

Reliance on distributed diesel generation as an alternative to stable grid-connected energy is common across Sub-Saharan Africa, these insights are therefore likely applicable more broadly across the region. Reliance on diesel generation extends outside of Africa with high volumes of generators relied upon globally but with particular prevalence in India, Lebanon, Thailand, Brazil, and Malaysia [7]. The particularities of the generation mix, decarbonisation goals, and grid in each of these countries will vary but the consideration of this reliance in any decarbonisation strategy, as broken down in this modelling, is crucial. Future work could explore similar themes in other regional and global contexts.

6. Conclusions

Nigeria struggles with significant reliability issues in its power grid which have spurred the dominance of diesel backup generation in the nation. These reliability issues can be partially traced back to low plant availability and insufficient capacity to meet growing demand. The

resulting persistent blackouts and reliance on generation at such a scale has marked impacts on the economic and social development of the nation and presents a considerable barrier in realising the latest round of NDCs.

Modelling has shown that lack of plant improvement, the consequent supply inadequacy, and reliance on diesel generation present significant barriers towards emissions reduction. Across scenarios, improved plant availability was found to lower emissions, system costs, and loss of load in the flexibility modelling. Increasing plant availability by 2 % each year until 2050, the more modest and achievable of the two improvement pathways, produces average emissions and cost savings of 45 % across scenarios.

These findings underscore the importance of improvements in plant and grid reliability for meeting emissions and development goals. They suggest that Nigeria should prioritise preventative maintenance and performance-reflective tariff structures as well as concrete energy efficiency standards/incentives in policies to support the 2021 NDCs.

CRedit authorship contribution statement

Sarah Golobish: Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Rudolf Yeganyan:** Writing – review & editing. **Naomi Tan:** Data curation. **Carla Cannone:** Software, Funding acquisition. **Mark Howells:** Software, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Sarah Golobish reports financial support and article publishing charges were provided by Loughborough University Climate Compatible Growth Programme. Co-author serves as an editor-in-chief for Energy Strategy Reviews - MH If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data and models used in this paper are openly available and are included in the repository below. <https://zenodo.org/doi/10.5281/zenodo.7783557>.

References

- [1] Nigerian Electricity Regulatory Commission, Key operational & financial data of NESI for Jan 2019 to Sept 2021. <https://nerc.gov.ng/index.php/component/remote/Industry-Statistics/Key-Operational-and-Financial-Data-of-NESI-for-Jan-2019-to-Sept-2021/>, 2021.
- [2] H.A. Daggash, N. Mac Dowell, Delivering low-carbon electricity systems in sub-Saharan Africa: insights from Nigeria, *Energy Environ. Sci.* 14 (2021) 4018–4037, <https://doi.org/10.1039/D1EE00746G>.
- [3] A.O. Soyemi, I.A. Samuel, A. Adesanya, A. Akinmeji, F. Adenugba, A robust energy policy review of selected African countries: an impetus for energy sustainability in

- Nigeria, J. Phys.: Conf. Ser. 1734 (2021) 012028, <https://doi.org/10.1088/1742-6596/1734/1/012028>.
- [4] S.O. Oyedepo, Energy and sustainable development in Nigeria: the way forward, *Energ Sustain Soc* 2 (2012) 15, <https://doi.org/10.1186/2192-0567-2-15>.
 - [5] IEA, IRENA, UNSD, World Bank, WHO, tracking SDG 7: the energy progress report. <https://data.worldbank.org/indicator/EG.ELC.ACCS.ZS?locations=NG>, 2023.
 - [6] M.O. Oseni, Get rid of it: to what extent might improved reliability reduce self-generation in Nigeria? *Energy Policy* 93 (2016) 246–254, <https://doi.org/10.1016/j.enpol.2016.03.013>.
 - [7] IFC, The Dirty Footprint of the Broken Grid: Summary, International Finance Corporation, 2019, <https://www.ifc.org/wps/wcm/connect/2cd3d83d-4f00-4d42-9bdc-4afdc2f5dbc7/20190919-Full-Report-The-Dirty-Footprint-of-the-Broken-Grid.pdf?MOD=AJPERES&CVID=mR9UpXC>.
 - [8] Power for all, Powering Jobs Census 2019: Focus on Nigeria, Power for All, 2019, <https://www.powerforall.org/resources/reports/powering-jobs-census-2019-focus-nigeria>.
 - [9] Federal Government of Nigeria, Nigeria's Nationally Determined Contribution – 2021 Update, Federal Government of Nigeria, Abuja, 2021. <https://unfccc.in/documents/497790>.
 - [10] Transmission Company of Nigeria, Electricity grid milestones. <https://www.niggrid.org/GenerationLoadProfileBinary>, 2022.
 - [11] Nigerian Electricity Regulatory Commission, NESI generation statistics. <https://nerc.gov.ng/index.php/component/remository/Industry-Statistics/NESI-Generation-Statistics/?Itemid=591>, 2018.
 - [12] S.O. Oyedepo, R.O. Fagbenle, S.S. Adefila, S.A. Adavbiele, Performance evaluation and economic analysis of a gas turbine power plant in Nigeria, *Energy Convers. Manag.* 79 (2014) 431–440, <https://doi.org/10.1016/j.enconman.2013.12.034>.
 - [13] SERAP (Socio-Economic Rights and Accountability Project), From darkness to darkness: how Nigerians are paying the price for corruption in the electricity sector. <https://serap-nigeria.org/download/from-darkness-to-darkness-how-nigerians-are-paying-the-price-for-corruption-in-the-electricity-sector/>, 2017.
 - [14] P. Roy, M. Watkins, C.K. Iwuamadi, J. Ibrahim, Breaking the cycle of corruption in Nigeria's electricity sector: off-grid solutions for local enterprises, *Energy Res. Social Sci.* 101 (2023) 103130, <https://doi.org/10.1016/j.erss.2023.103130>.
 - [15] L. Ekeng, C. Ahiakwo, H. Amadi, E. Obuah, Voltage collapse in Nigeria power System- causes and remedies, (n.d.). <https://doi.org/10.9790/0853-1901015463..>
 - [16] U. Justin, K. Oseni, L. Onyedikachi, O.S. Ejiofor, Incidences of Voltage Collapse in the Nigerian Power System: Data and Analysis, vol. 63, 2021, p. 19.
 - [17] Advisory Power Team, Power Africa, Nigeria Power Baseline Report, Office of the Vice President, Federal Government of Nigeria, 2015. <https://mypower.ng/wp-content/uploads/2018/01/Baseline-Report.pdf>.
 - [18] FMITI, Nigeria Industrial Revolution Plan, Federal Ministry for Industry Trade and Investment, 2014. <https://nipc.gov.ng/wp-content/uploads/2019/03/nirp.pdf>.
 - [19] D. Farquharson, Sustainable energy transitions in Sub-Saharan Africa: impacts on air quality, economics, and fuel consumption. <https://www.cmu.edu/ceic/research-publications/devynne-farquharson-phd-thesis-2019.pdf>, 2019.
 - [20] Federal Government of Nigeria, Nigeria energy transition plan. <https://www.energytransition.gov.ng>, 2022.
 - [21] N. Avila, J.P. Carvallo, B. Shaw, D.M. Kammen, *The Energy Challenge in Sub-Saharan Africa: a Guide for Advocates and Policy Makers*, 2017, p. 79.
 - [22] H. Ibrahim, G. Kirkil, Electricity demand and supply scenario analysis for Nigeria using long range energy alternatives planning (LEAP), *JSRR* 19 (2018) 1–12, <https://doi.org/10.9734/JSRR/2018/39719>.
 - [23] J.O. Dada, A. Moser, REPLAN: Multi-region power system planning approach for Nigeria, in: 2019 IEEE PES/IAS PowerAfrica, IEEE, Abuja, Nigeria, 2019, pp. 87–92, <https://doi.org/10.1109/PowerAfrica.2019.8928896>.
 - [24] IRENA, IRENA FlexTool: summary of methodology. https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2018/Nov/IRENA_FlexTool_summary_2019.pdf?la=en&hash=046D20164074C7BB9D6A18351F35050875963014, 2019.
 - [25] Nigerian Electricity Regulatory Commission, Nigeria Electricity Regulatory Commission Quarterly Report: Fourth Quarter 2020, Nigeria Electricity Regulatory Commission, 2020. <https://nerc.gov.ng/index.php/library/documents/func-start-down/901/>.
 - [26] Global Energy Monitor, Global power plant tracker. <https://globalenergymonitor.org>, 2022.
 - [27] Federal government of Nigeria, sustainable energy for all action agenda, Federal Government of Nigeria. https://www.seforall.org/sites/default/files/NIGERIA_S_E4ALL_ACTION_AGENDA_FINAL.pdf, 2016.
 - [28] REA, The Off-Grid Opportunity in Nigeria, Rural Electrification Agency, 2017. https://www.esmap.org/sites/default/files/Presentations/REA_Damiloia-Off-Grid%20Opportunity_03122017_web.pdf.
 - [29] M.O. Dioha, A. Kumar, Exploring sustainable energy transitions in Sub-Saharan Africa residential sector: the case of Nigeria, *Renew. Sustain. Energy Rev.* 117 (2020) 109510, <https://doi.org/10.1016/j.rser.2019.109510>.
 - [30] L. Allington, C. Cannone, I. Pappis, K.C. Barron, W. Usher, S. Pye, M. Howells, M. Z. Walker, A. Ahsan, F. Charbonnier, C. Halloran, S. Hirmer, C. Taliotis, C. Sundin, V. Sridharan, E. Ramos, M. Brinkerink, P. Deane, A. Gritsevskiy, G. Moura, A. Rouget, D. Wogan, E. Barcelona, H. Rogner, N. Tan, CCG starter data kit: nigeria. <https://doi.org/10.5281/ZENODO.6542376>, 2022.
 - [31] IRENA, West African Power Pool, Planning and prospects for renewable energy, International Renewable Energy Agency, <https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2013/WAPP.pdf>, 2013.
 - [32] C. Cannone, L. Allington, K.C. Barron, F. Charbonnier, M.Z. Walker, C. Halloran, R. Yeganyan, N. Tan, J.M. Cullen, J. Harrison, L.S. To, M. Howells, Designing a zero-order energy transition model: how to create a new Starter Data Kit, *MethodsX* 10 (2023) 102120, <https://doi.org/10.1016/j.mex.2023.102120>.
 - [33] M. Brinkerink, P. Deane, PLEXOS-World 2015. <https://doi.org/10.7910/DVN/CBYXBY>, 2020.
 - [34] E. Taibi, T. Nikolakakis, L. Gutierrez, C. Fernandez, J. Kiviluoma, S. Rissanen, T. J. Lindroos, Power system flexibility for the energy transition: Part 2, IRENA FlexTool methodology, 2018.
 - [35] L. Allington, C. Cannone, I. Pappis, K.C. Barron, W. Usher, S. Pye, E. Brown, M. Howells, M.Z. Walker, A. Ahsan, F. Charbonnier, C. Halloran, S. Hirmer, C. Taliotis, C. Sundin, V. Sridharan, E. Ramos, E. Ramos, P. Deane, A. Gritsevskiy, G. Moura, A. Rouget, D. Wogan, E. Barcelona, H. Rogner, Selected 'Starter Kit' energy system modelling data for Nigeria (#CCG). <https://doi.org/10.21203/rs.3.rs-480085/v2>, 2021.
 - [36] H.N. Amadi, *Power Outages in Port Harcourt City: Problems and Solutions, Problems and Solutions*, 2015.
 - [37] B. Akintunde, COP26: nigeria will cut carbon emission to net zero by 2060, Buhari says, *Times Nigeria*. <https://www.premiumtimesng.com/news/headlines/493040-cop26-nigeria-will-cut-carbon-emission-to-net-zero-by-2060-buhari-says.html>, 2021.
 - [38] UNDESA, 2030 agenda for sustainable development. <https://sdgs.un.org/goals>, 2022.
 - [39] A. Babajide, M.C. Brito, Solar PV systems to eliminate or reduce the use of diesel generators at no additional cost: a case study of Lagos, Nigeria, *Renew. Energy* 172 (2021) 209–218, <https://doi.org/10.1016/j.renene.2021.02.088>.
 - [40] A. Ambole, K. Koranteng, P. Njoroge, D.L. Luhangala, A review of energy communities in Sub-Saharan Africa as a transition pathway to energy democracy, *Sustainability* 13 (2021) 2128, <https://doi.org/10.3390/su13042128>.